CREDIT | GUARD

AI-POWERED CREDIT RISK ANALYSIS

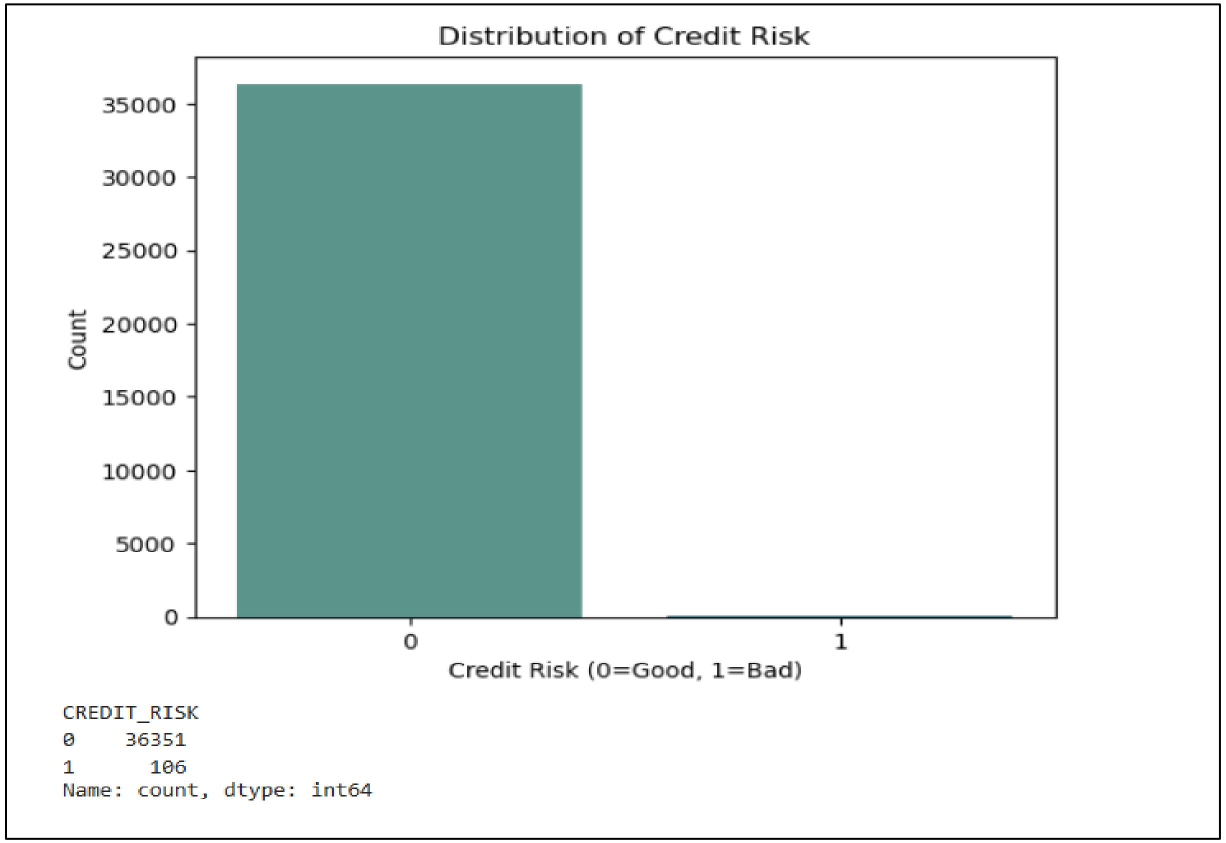
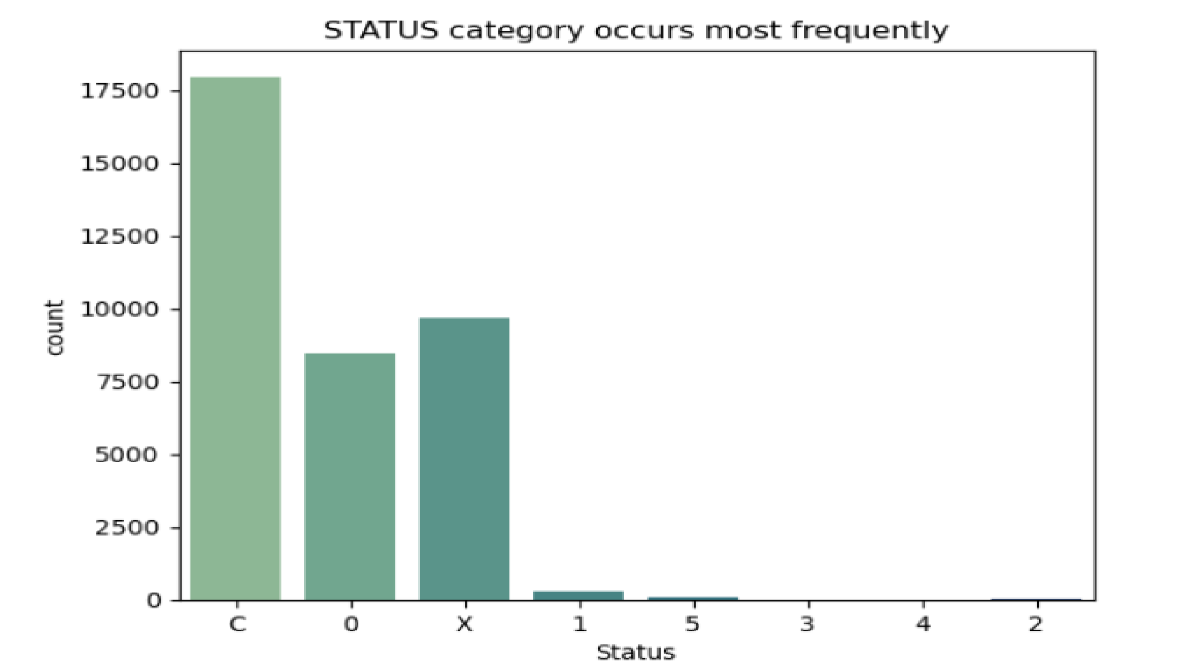


FIG 1. Distribution of Credit Risk

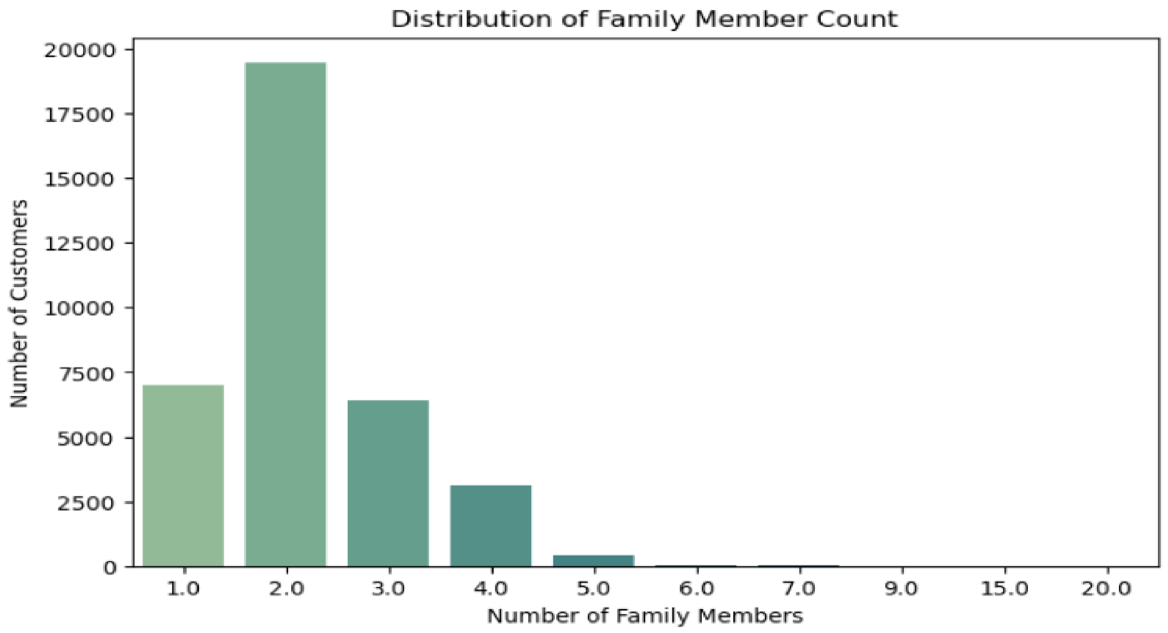
- The **CREDIT_RISK** variable is **highly imbalanced**, with a significant majority of customers classified as **Good Credit (0)**.
- Out of **36,457** customers, **36,351 (99.71%)** are classified as **Good Credit**, while only **106 (0.29%)** are classified as **Bad Credit**.
- The extremely small number of bad credit cases indicates that **defaults are rare** in this dataset.
- This class imbalance may lead to **biased machine learning models**, as models can become biased toward predicting the majority class.
- Special techniques such as **SMOTE, oversampling, undersampling, or class weighting** may be required during model training to handle the imbalance.
- Evaluation metrics such as **Precision, Recall, F1-score, and ROC-AUC** should be preferred over accuracy alone when assessing model performance on this dataset.




```
STATUS
C    17950
X     9669
0     8456
1      276
5       76
2       19
3        6
4        5
Name: count, dtype: int64
```

FIG 2. Status category occurs most frequently

- The most common repayment status is **C (Loan Closed)** with **17,950 customers (49.24%)**, followed by **X (No Loan Activity)** with **9,669 customers (26.52%)**.
- **8,456 customers (23.19%)** have a status of **0 (Paid on Time)**, indicating regular and timely repayments.
- Only **382 customers (1.05%)** fall into the late payment or default categories (1–5), showing that delinquent repayments are uncommon.
- The low frequency of severe delinquency (2–5) suggests that **most customers have a stable repayment history**.
- The dominance of **C**, **X**, and **0** indicates that the dataset consists primarily of customers with **low credit risk**.
- The scarcity of late payment statuses further supports the **highly imbalanced distribution of the target variable (CREDIT_RISK)**, where bad credit cases are very rare.




```
CNT_FAM_MEMBERS
2.0      19463
1.0       6987
3.0       6421
4.0       3106
5.0        397
6.0         58
7.0         19
15.0         3
9.0          2
20.0         1
Name: count, dtype: int64
```

FIG 3. Distribution of Family Member Count

- The majority of customers have **2 family members (19,463; 53.39%)**, making it the most common family size.
- Customers with **1** and **3 family members** are the next largest groups, indicating that **small families dominate the dataset**.
- Family sizes of **4 or 5 members** are less common but still reasonably represented.
- Very few customers have **more than 5 family members**, suggesting that large families are rare.
- A handful of records with **9, 15, and 20 family members** may represent **extreme values or potential outliers** that warrant further investigation.
- Overall, the distribution indicates that **most customers belong to small or medium-sized households**.

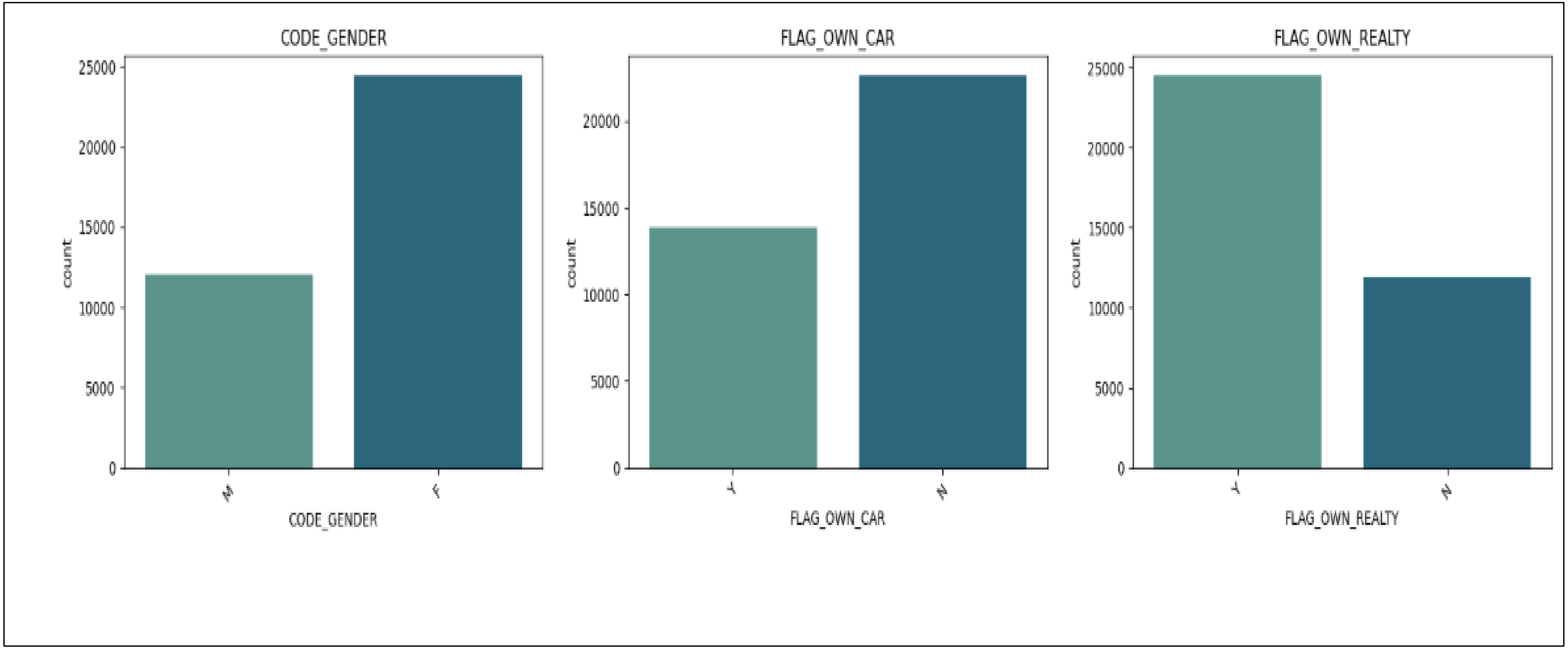


FIG 4. Value counts of Gender,Owning car,Owning real estate

- The dataset contains **more female customers than male customers**, indicating that females constitute the majority of applicants.
- A larger proportion of customers **do not own a car (N)** compared to those who own a car (Y).
- Most customers **own a house or property (Y)**, suggesting that property ownership is common among the applicants.
- The higher percentage of property owners may indicate **greater financial stability**, which could positively influence creditworthiness.
- Although car ownership is less common, **property ownership is significantly higher**, indicating that owning real estate is more prevalent than owning a car.
- These demographic and asset ownership characteristics can be useful features for analyzing customer profiles and their relationship with **credit risk**.

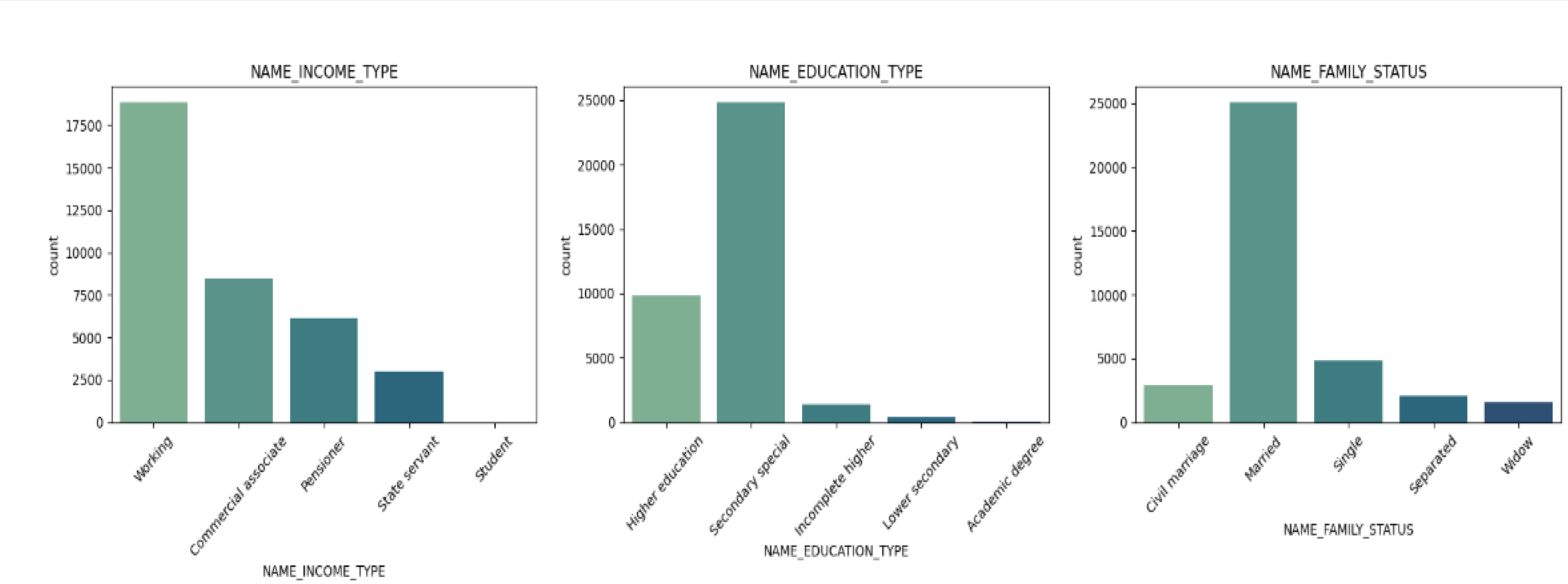


FIG 5. Value counts of Income type, Education type, Family Status

- The majority of customers are **Working**, followed by **Commercial Associates** and **Pensioners**, while **Students** represent only a negligible proportion of the dataset.
- Most customers have **Secondary/Special Secondary Education**, followed by **Higher Education**, indicating that the dataset primarily consists of customers with basic to intermediate educational qualifications.
- Customers with **Academic Degrees** and **Lower Secondary Education** are very few compared to other education levels.
- The majority of customers are **Married**, making it the most common family status in the dataset.
- **Single** customers form the second-largest group, followed by **Civil Marriage**, **Separated**, and **Widow** categories.
- The distributions suggest that the typical customer profile is a **working, married individual with secondary-level education**.
- These demographic characteristics may influence customers' financial behavior and can be valuable predictors in **credit risk analysis**.

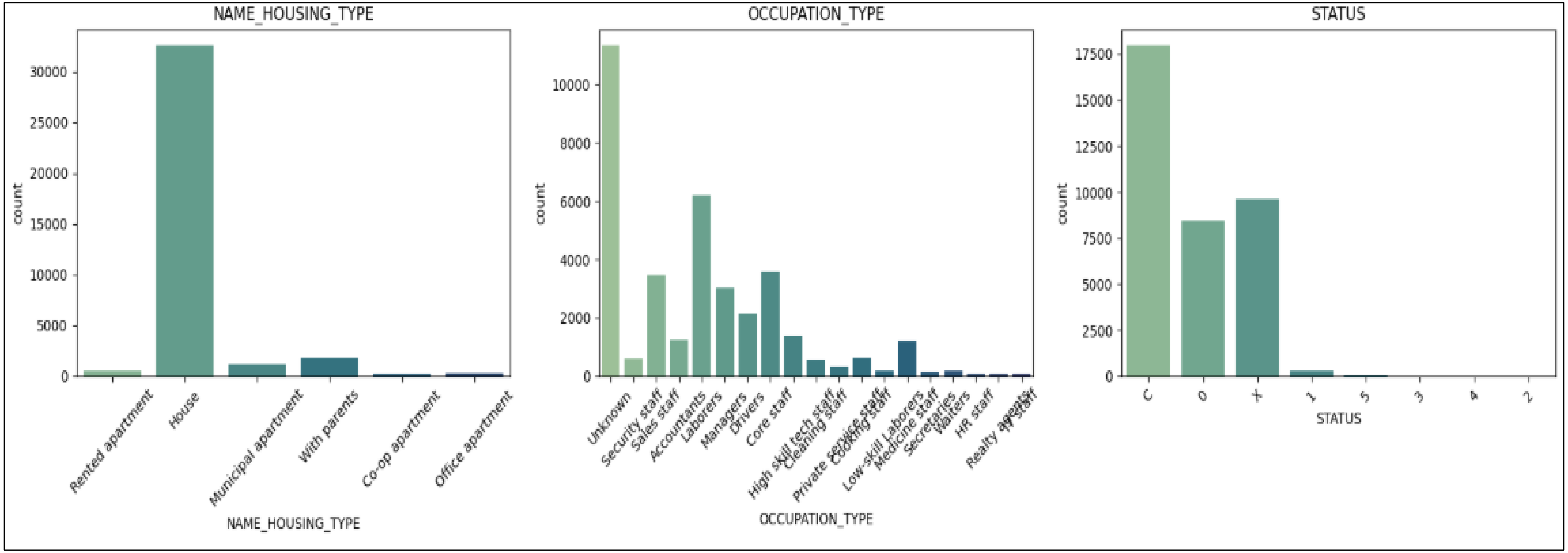


FIG 6. Value counts of Housing type,Occupation type, Status

- The majority of customers **live in their own house**, while very few reside in rented apartments, office apartments, or co-op apartments.
- A small proportion of customers **live with their parents** or in **municipal apartments**, indicating that independent home ownership is common.
- The **Unknown** occupation category has the highest number of customers, followed by **Laborers**, **Core Staff**, and **Sales Staff**.
- Occupations such as **HR Staff, IT Staff, Secretaries, Realty Agents,** and **Waiters** have relatively few customers in the dataset.
- The **C (Loan Closed)** status is the most frequent repayment status, followed by **X (No Loan Activity)** and **0 (Paid on Time)**.
- Very few customers fall into the **late payment or default categories (1–5)**, indicating that serious repayment issues are uncommon.
- Overall, the dataset consists mainly of **homeowners with stable repayment histories**, while severe credit delinquency is observed only in a small fraction of customers.

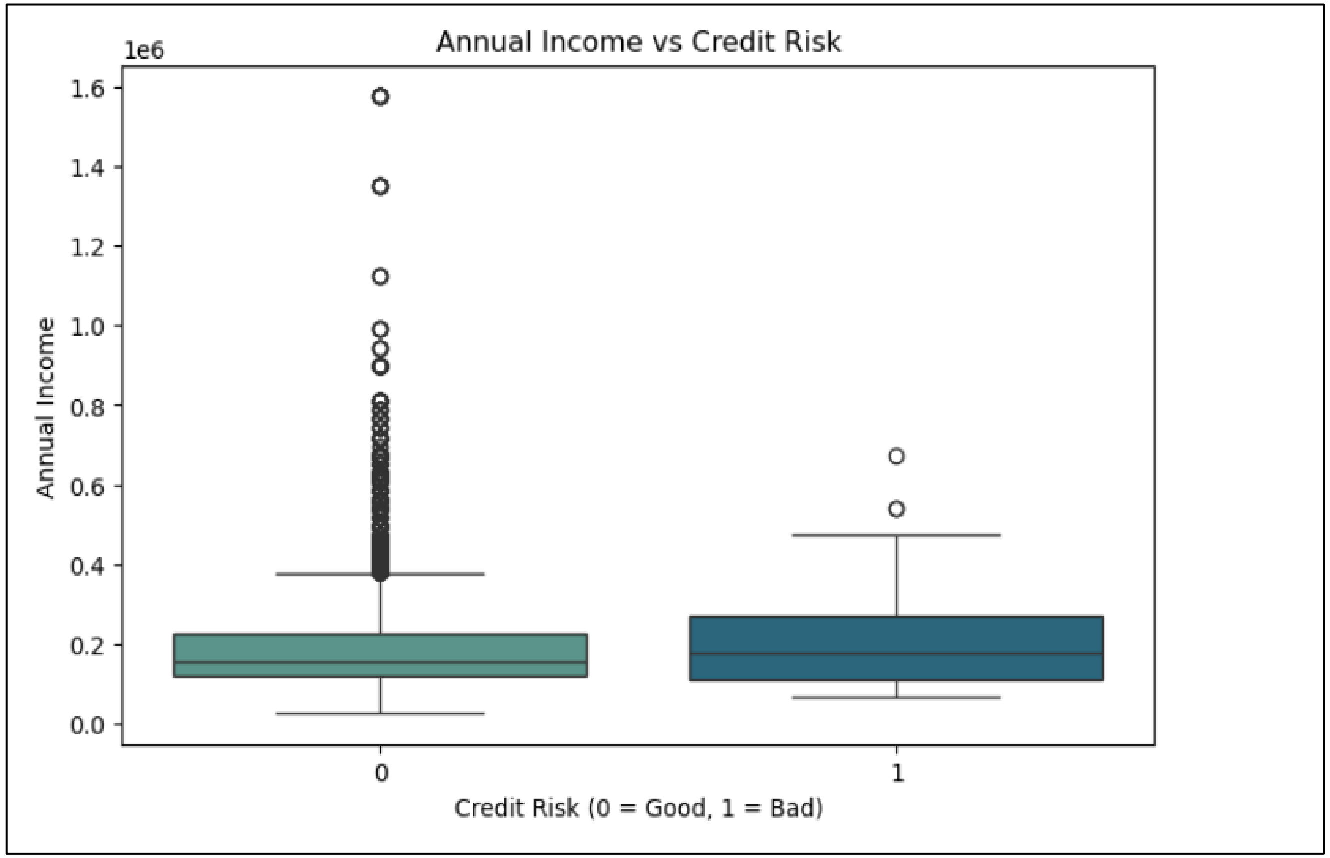


FIG 7. Annual Income VS Credit Risk

- The median annual income of **bad credit customers (1)** is **slightly higher** than that of **good credit customers (0)**.
- The annual income distributions of both groups **overlap considerably**, suggesting that income alone does not clearly distinguish credit risk.
- The **good credit group (0)** contains a large number of **high-income outliers**, indicating greater variability in income.
- The **bad credit group (1)** has relatively **fewer outliers**, likely due to the small number of bad credit customers in the dataset.
- Most customers in both credit risk groups have **annual incomes within a similar range**, with only a few earning exceptionally high incomes.
- Overall, **annual income does not appear to have a strong relationship with credit risk** in this dataset, as both groups exhibit comparable income distributions.

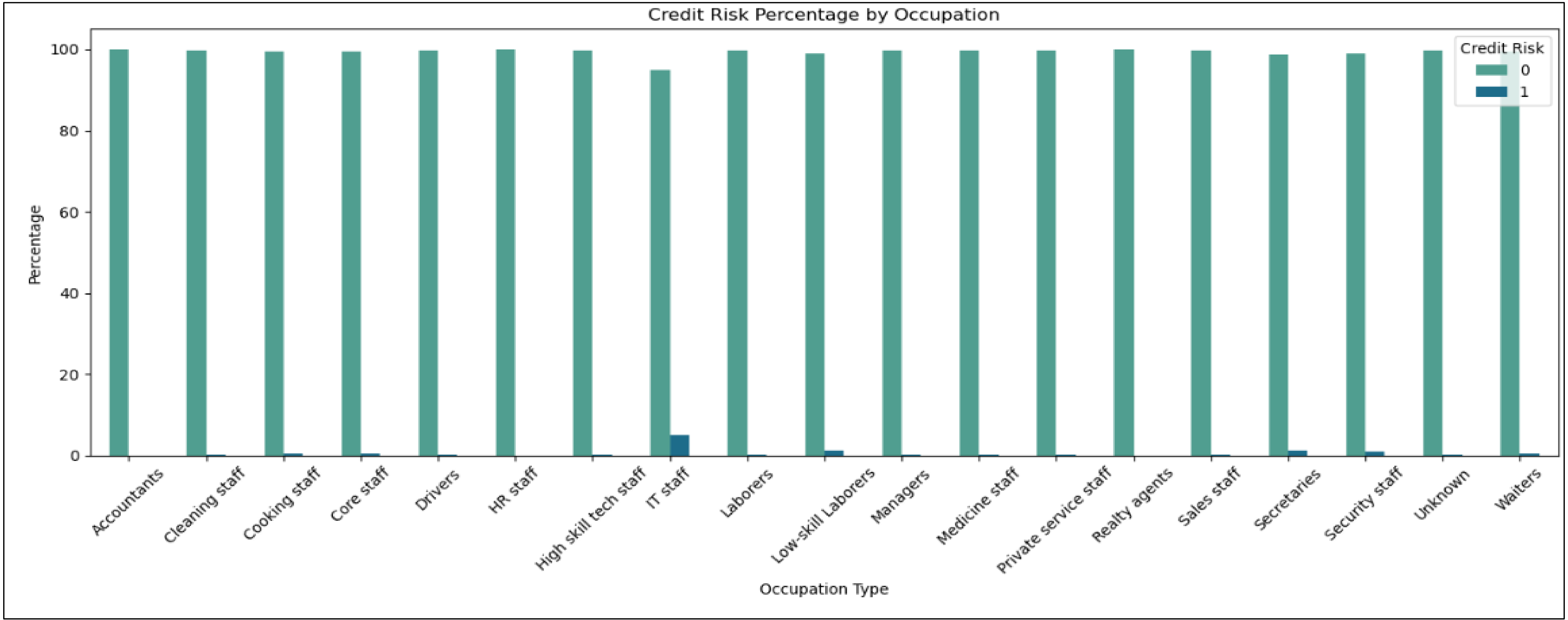
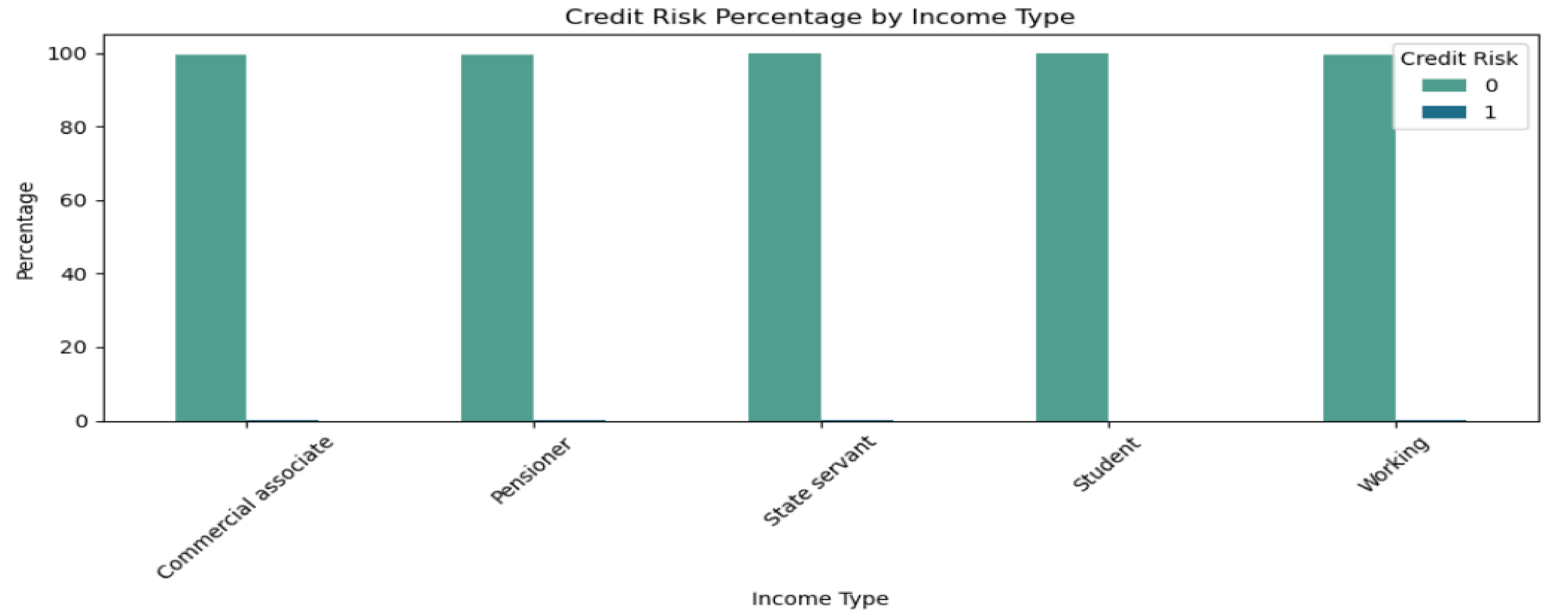


FIG 8. Credit Risk Percentage by Occupation

- Across all occupation types, **more than 98% of customers have good credit (Credit Risk = 0)**, indicating a generally low default rate.
- **Secretaries (1.32%)**, **Low-skill Laborers (1.14%)**, and **Security Staff (1.01%)** have the highest proportions of bad credit customers among all occupations.
- Occupations such as **Accountants**, **HR Staff**, and **Realty Agents** show **100% good credit**, with no bad credit cases observed.
- The **IT Staff** category has a relatively high bad credit rate (**5%**), but this result should be interpreted cautiously because it is likely based on a **very small number of customers**.
- The **Unknown** occupation category also has a very high proportion of good credit customers (**99.75%**), similar to most other occupations.
- Overall, **occupation type alone does not appear to be a strong predictor of credit risk**, as bad credit cases are extremely rare across all occupation categories.



CREDIT_RISK	0	1
NAME_INCOME_TYPE		
Commercial associate	99.646643	0.353357
Pensioner	99.658648	0.341352
State servant	99.932998	0.067002
Student	100.000000	0.000000
Working	99.718370	0.281630

FIG 9. Credit Risk Percentage by Income Type

- All income types have an **exceptionally high proportion of good credit customers (above 99%)**, indicating a very low overall default rate.
- **Commercial Associates (0.35%)** and **Pensioners (0.34%)** have the highest percentages of bad credit customers among the income groups.
- **Working** customers also exhibit a very low bad credit rate (**0.28%**), suggesting stable repayment behavior.
- **State Servants** have one of the lowest bad credit rates (**0.07%**), indicating excellent credit performance.
- **Students** show **100% good credit**, with no bad credit cases observed; however, this result may be due to the very small number of student records.
- Overall, **income type does not appear to have a strong influence on credit risk**, as good credit customers dominate every income category.